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NAAC GRADE A+

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ASSIGNMENT-

Subject: Database Management System

Subject Code: 24CSH-298

Branch Sec: 24AIT KRG1-G2

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Submitted To:

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Q1) Types of Security Issues. Explain the following types of database security issues: 1. Legal and Ethical Issues
2. Policy Issues
3. System-related issues

Ans: Types of Database security Issues:

1) Legal and Ethical Issue

These issues are related to laws, regulations and ethical responsibilities for protecting data. Organizations must follow data protection laws and ensure that personal or confidential data is not misused.

Examples:

- Protecting user privacy
- Avoiding unauthorized sharing of data
- Following data protection regulations

2) Policy Issues

Policy issues refer to rules and guidelines set by an organization to manage database security

These policies define:

- Who can access the database
- What actions users are allowed to perform
- How data should be protected.

Example: Only managers can access salary information.

3> System - related Issues

These issues occur due to technical weaknesses in the database System.

Example include :

- Weak passwords
- Software vulnerabilities
- Unauthorized access to the database
- Malware or hacking attacks

Proper System security measures must be used to prevent these problems.

Q2> Database Security Control Techniques

Explain the following types of database security ~~issues~~:

Techniques : 1> Access Control 2> Inference Control 3> Flow Control 4> Data Encryption

Ans: 1> Access Control

Access control restricts who can access the database and what operations they can perform.

Examples

- User Authentication (login with username and password)
- Role - based access (admin, user, manager)

2> Inference Control

Inference control prevents users from deducing sensitive information from available data.

Example :

Confidential data should not be accessed by unauthorized users.

3) Flow Control :

Flow control ensures that information does not flow from higher security levels to lower security levels.

Example :

Confidential data should not be accessed by unauthorized users.

4) Data Encryption

Data encryption converts data into coded form so that unauthorized users cannot understand it.

Example :

- Password encryption
- Secure data transmission.

Encryption protects data during storage and transmission.

Q3) SQL View Creation

Consider a table Employee ~~ID~~ (Emp ID, Emp Name, Salary, Dept ID). Write an SQL query to create a view that displays only Emp ID, Emp Name, and Dept ID from the Employee table. give me answers for my assignment.

Ans: SQL View Creation

Table:

Employee (EmpID, EmpName, Salary, DeptID)

SQL Query

```
CREATE VIEW EMployee_View AS
SELECT EmpID, EmpName, DeptID
FROM Employee;
```

Explanation

This query creates a view named Employee_View that display only:

- EmpID
- EmpName
- Dept ID

The Salary column is hidden in this view.